

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 (R \otimes 1_L)(\alpha \otimes \varphi_L)\Delta(l) &= (R \otimes 1_L)(\alpha \otimes \varphi_L)(l_1 \otimes l_2) \\
 &= (R \otimes 1_L)(\alpha(l_1) \otimes \varphi_L(l_2)) \\
 &= R(\alpha(l_1)) \otimes \varphi_L(l_2) \\
 &\stackrel{(??.)}{=} \alpha(R(l_1)) \otimes \varphi_L(l_2) \\
 &= (\alpha \otimes \varphi_L)(R \otimes 1_L)\Delta(l).
 \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 I_\mu(u) &= \frac{1}{2} \int_{\mathbb{R}^N} (|\nabla u|^2 + (\mu + 1)u^2) dx + \int_{\mathbb{R}^N} F_1(u) dx - \int_{\mathbb{R}^N} F_2(u) dx \\
 &\geq \frac{1}{2} \int_{\mathbb{R}^N} |\nabla u|^2 dx + \int_{\mathbb{R}^N} F_1(u) dx - CC_1 a^{(1-\beta_p)p} \left(\int_{\mathbb{R}^N} |\nabla u|^2 dx \right)^{\frac{p\beta_p}{2}}.
 \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 a_{ij}^{\mathbf{4}} &= \frac{1}{g} \sum_{\gamma=1}^r r_\gamma \chi_\gamma^{R_M \oplus \mathbf{1}^2} \chi_\gamma^{(i)} \chi_\gamma^{(j)*} \\
 a_{ij}^{\mathbf{6}} &= \frac{1}{g} \sum_{\gamma=1}^r r_\gamma \chi_\gamma^{R_M \oplus R_M \oplus \mathbf{1}^2} \chi_\gamma^{(i)} \chi_\gamma^{(j)*}
 \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
 (\tilde{R}_\Sigma(\eta)E)^{s-1}P_rE(R_\Sigma(\eta)E)^{k-s}P_rv &= \sum_{l \in \Delta_r} (\tilde{R}_\Sigma(\eta)E)^{s-1}(\theta_l \otimes \theta_l)E(R_\Sigma(\eta)E)^{k-s}P_rv = \\
 &= \sum_{l \in \Delta_r} \left\langle E(R_\Sigma(\eta)E)^{k-s}P_rv, \theta_l \right\rangle (\tilde{R}_\Sigma(\eta)E)^{s-2}\tilde{R}_\Sigma(\eta)E\theta_l.
 \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
sA_\mu^a &= -(D_\mu c)^a \quad , \\
\begin{array}{l} \text{beginalign*2mm} \end{array} s\phi^a &= f^{abc} c^b \phi^c \quad , \\
\begin{array}{l} \text{beginalign*2mm} \end{array} sc^a &= \frac{1}{2} f^{abc} c^b c^c \quad , \\
\begin{array}{l} \text{beginalign*2mm} \end{array} s\bar{c}^a &= \bar{b}^a \quad , \\
\begin{array}{l} \text{beginalign*2mm} \end{array} sb^a &= 0 \quad , \\
\begin{array}{l} \text{beginalign*2mm} \end{array} s^2 &= 0 \quad .
\end{aligned}$$